

Event-Driven Tactile Object Classification with Spiking Neural Networks and Input Compression

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Abstract—Tactile sensing provides rich information for prosthetic and robotic manipulation, but its high dimensionality challenges compact event-driven processing and neuromorphic deployment. We investigate how tactile data can be transformed into compact event-driven representations for object classification through the systematic study of input compression and Spiking Neural Networks (SNNs).

Several spatial compression strategies, including PCA-based selection, farthest point sampling, and spatial averaging, are evaluated on the ETHZ-STAG dataset to preserve classification performance with compact representations. The resulting spike-based inputs are classified using a fully connected SNN, showing competitive performance and improved inter-session robustness compared with a previously reported CNN. Finally, proof-of-concept deployment on a neuromorphic processor demonstrates operation under real mixed-signal constraints.

Overall, the results demonstrate the feasibility of end-to-end neuromorphic tactile object classification and provide practical design insights into spatial compression strategies for future robotic and prosthetic systems.

I. INTRODUCTION

Tactile sensing is essential for dexterous manipulation, enabling the perception of contact forces, texture, slip, and vibration through mechanoreceptors that encode information such as spike trains [1], [2]. In prosthetics, the absence of this feedback limits usability, forcing reliance on visual signals and increasing cognitive load, often leading to reduced precision and device abandonment [3]. Similarly, in robotics, the lack of reliable tactile feedback restricts manipulation, particularly in tasks requiring fine force control and interaction with uncertain environments [4], [5].

Recent advances in electronic skins provide high-resolution multimodal sensing, but conventional systems rely on frame-based sampling and centralized processing, resulting in high computational and communication demands, increased latency, and poor scalability [6], [7], [8]. Moreover, they fail to capture the fast temporal dynamics that are critical for slip and texture perception. Bio-inspired approaches address these limitations by generating spike-based tactile signals compatible with neuromorphic processing, enabling sparse event-driven computation [9], [10]. However, many systems still rely on streaming high-bandwidth data to external processors, limiting scalability and real-time applicability.

Neuromorphic computing offers a promising alternative by processing sparse asynchronous events directly. Spiking Neural Networks (SNNs) process information through discrete

spikes, enabling sparse temporal computation that is well suited to time-varying signals. They have demonstrated strong performance in biomedical signal processing [11], [12], where temporal dynamics play a central role [13], [14], and in tactile applications including texture recognition, slip detection, and Braille classification [15], [16], [17]. Their event-driven nature makes them particularly attractive for tactile signals such as grasp onset, slip, and force modulation [14], [18], [19].

Object classification using tactile sensing provides a relevant benchmark task, directly supporting grasp control and manipulation. The SmartHand platform [20] enables high-resolution tactile acquisition with a 32×32 sensor array (1024 taxels) at 100 Hz, forming the ETHZ-STAG dataset. Prior work demonstrated strong performance using Convolutional Neural Networks (CNNs), achieving up to 98.86% accuracy under random splits and 77.84% top-3 accuracy under session-based evaluation [20]. However, these approaches rely on frame-based processing and continuous sampling, limiting energy efficiency and underutilizing temporal dynamics.

To address these challenges, we systematically investigate how different spatial compression strategies influence the quality of event-driven tactile representations and their impact on classification performance within a complete neuromorphic processing pipeline comprising ADM encoding, SNN-based classification, and proof-of-concept deployment on the DYNAP-SE neuromorphic processor [21]. This analysis identifies the trade-offs between input dimensionality and classification performance, providing practical design guidelines for compact event-driven tactile representations suitable for future neuromorphic tactile systems. Overall, the results demonstrate the feasibility of end-to-end neuromorphic tactile object classification for robotic and prosthetic applications.

II. MATERIAL & METHODS

A. Dataset

The ETHZ-STAG dataset [20], [22], building on the MIT-STAG dataset [23], was collected using the SmartHand system for tactile sensing in prosthetic and robotic applications. It contains recordings of 16 everyday objects and an empty-hand baseline, for a total of 17 classes. Data were acquired across five sessions while a participant repeatedly grasped and released each object using a tactile glove with a 32×32 layout (1024 taxels), of which 548 correspond to valid electrode crossings on the hand-shaped sensing region. Signals were

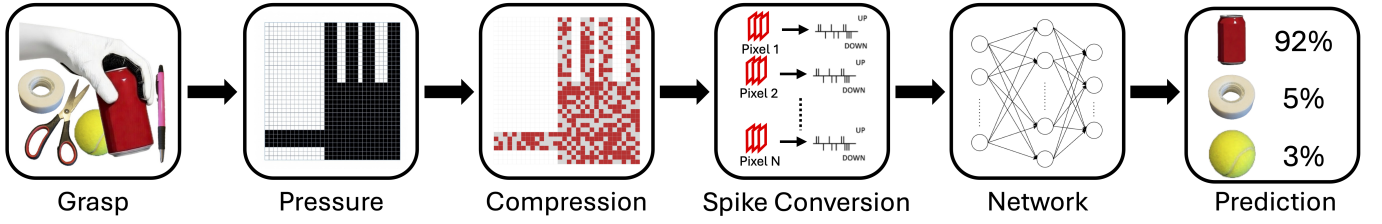


Fig. 1: Tactile-based object classification pipeline using pressure sensing and spiking neural networks. From left to right: multi-object grasp, pressure map, compressed representation, spike encoding, neural network, and final prediction.

sampled at 100 Hz, yielding approximately 340,000 frames. Hyperparameter tuning and compression studies used sessions 0 and 1 as a fixed development subset, while inter-session generalization was evaluated across all five sessions using leave-one-session-out cross-validation, with each session held out in turn for evaluation.

B. Tactile Data Preprocessing

Tactile values and baseline thresholds are first normalized independently to $[0, 1]$, after which baseline subtraction is performed. Frames marked as invalid are discarded. This preprocessing reduces sensor offsets and inter-session variability while keeping the signal within a bounded range suitable for spike encoding.

The dataset is then balanced per session-class pair by retaining up to a fixed number of consecutive frames from each object and session, ensuring a more uniform distribution across the 17 classes and five sessions. Finally, data are grouped into temporal windows of T frames (e.g. $T \in \{25, 50\}$) corresponding to 250 ms or 500 ms at 100 Hz).

C. Input Signal Compression Methods

Tactile inputs consist of high-dimensional pressure maps with significant spatial redundancy due to correlations between neighboring sensors. Reducing dimensionality decreases the number of input channels processed by the classifier, reduces overfitting, and facilitates the extraction of informative features. Several methods are considered to reduce the effective input dimensionality while preserving discriminative information.

PCA-based Correlation Ranking (PCA-rank) (Fig. 2a): Principal components analysis (PCA) is applied to the centered tactile data to estimate the intrinsic dimensionality of the 548 valid hand taxels. Only 98 components explain 90% of the variance, while 158 explain 95%, indicating substantial redundancy in the input space.

Farthest Point Sampling (Fig. 2b): Farthest Point Sampling (FPS) selects taxels iteratively by maximizing the minimum Euclidean distance to already selected points. Applied to the 548 valid taxels, it promotes uniform spatial coverage and avoids clustering.

Custom Masks (Fig. 2c): binary masks allow manual selection of taxels or regions of interest, enabling focus on specific areas such as fingertips or palm regions.

Spatial Averaging (Fig. 2d): the hand is partitioned into rectangular super-pixel regions with configurable kernel sizes. Taxels within each region are averaged, producing fewer inputs while preserving coarse spatial structure.

Figure 2 compares representative spatial selections produced by the main compression methods when retaining approximately the same number of inputs.

D. Spike Conversion

Centered pressure signals are converted into spike trains using Asynchronous Delta Modulator (ADM) [24], [25] (Fig. 3). Each taxel is encoded independently by comparing the signal to an adaptive baseline V_{DC} . UP and DOWN spikes are generated for positive and negative deviations beyond a threshold θ , with a 10 ms refractory period.

Events are represented as (x, t, p) , where x is the taxel index, t the timestamp, and $p \in \{+1, -1\}$ the polarity. After binning into 10 ms intervals, the data are represented either as (T, N) or $(T, 2N)$.

ADM encodes pressure variations as sparse events, with polarity indicating the direction of change and spike rate reflecting its magnitude. The representation is well suited to transient tactile events such as contact onset and slip.

E. SNN Architecture and Training

The SNN is a shallow feedforward network implemented in `snntorch` [26], comprising N spike-encoded tactile inputs, a fully connected hidden layer with $3N$ neurons, and 17 output neurons (Fig. 4). Hidden and output neurons include synaptic and membrane dynamics and emit spikes upon reaching threshold; classification is based on the output neuron with the highest spike count.

The network was trained using surrogate-gradient backpropagation through time [27] and Adam, with model selection based on validation performance. For each class and session, the first 1200 valid frames were retained; the last 100 were used for testing and the remainder for 5-fold cross-validation.

F. On-Chip Implementation

For proof-of-concept hardware deployment, the trained SNN was mapped to the DYNAP-SE neuromorphic processor [28]. The mapping was constrained by available neuron resources, a maximum fan-in of 64 synapses per neuron, and 2-bit synaptic-weight resolution. Accordingly, the input was compressed to $N = 128$ taxels and the task reduced to 10 classes, resulting

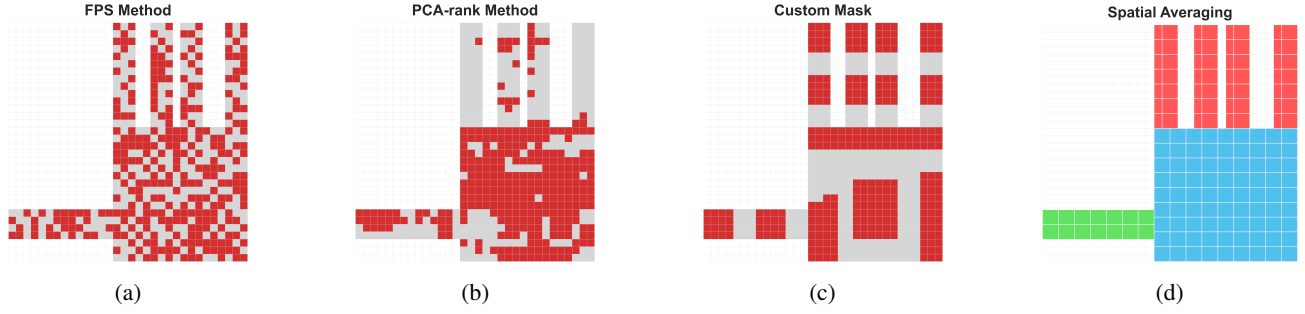


Fig. 2: Comparison of spatial selections produced by the input compression methods, each retaining approximately 300 active taxels (or super-pixels). Red indicates selected inputs to the SNN; white are ignored. (a) Farthest Point Sampling (FPS); (b) PCA-rank; (c) Custom mask; (d) Spatial averaging partitions.

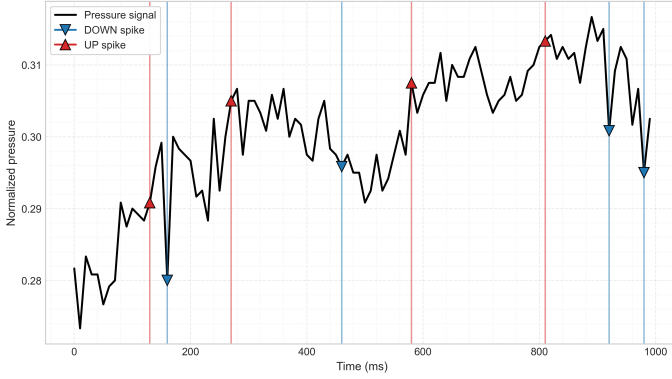


Fig. 3: ADM encoding example for a single taxel pressure signal over 100 frames (1000 ms at 100 Hz).

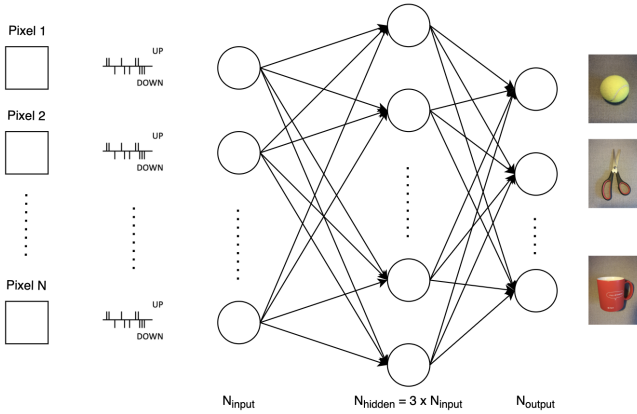


Fig. 4: SNN architecture with N spike-encoded tactile inputs, $3N$ hidden neurons, and 17 output neurons.

in 128 input spike generators, 384 hidden neurons, and 10 output neurons distributed across multiple cores with sparse first-layer connectivity.

Training was performed in a chip-in-the-loop setup: hidden and output spikes were recorded on chip, while readout weights were updated in software and written back to the device. Thus, inference was performed on chip, whereas learning

remained external. Only fast excitatory synapses were used, with neuron and synapse dynamics configured through global bias parameters. This setup enabled evaluation of the proposed tactile processing pipeline under the connectivity, resource, and mixed-signal constraints of the physical processor.

III. RESULTS

A hyperparameter sweep over window size, ADM threshold, and encoding strategy identified a 25-frame window, $\theta = 0.01$, and single-channel encoding with positive DOWN contribution as the best configuration, achieving 86.97% validation accuracy.

As shown in Table I, positive DOWN values provided the best performance, whereas negative values in the single-channel representation substantially degraded accuracy, likely due to interference between UP and DOWN events. This suggests that a compact additive representation is more effective than explicit polarity separation for this task.

TABLE I: Average validation accuracy across folds after 15 epochs for different encoding strategies.

Encoding	Avg. Val. Acc.
Double-channel (DOWN = -1.0)	81.12%
Double-channel (DOWN = 1.0)	80.67%
Single-channel (DOWN = -1.0)	20.45%
Single-channel (DOWN = 1.0)	86.97%

Input compression was then evaluated to determine how different spatial representations affect classification performance as input dimensionality is reduced. PCA-rank, FPS, random sampling, custom masks, and spatial averaging were compared (Table II). Among the automated approaches, FPS consistently provided the strongest performance, reaching 86.22% accuracy with 400 inputs and retaining 58.98% with only 100 inputs. At comparable dimensionality, FPS outperformed PCA-rank, suggesting that maintaining spatially distributed coverage of the tactile surface preserves discriminative contact information better than selecting taxels primarily according to global variance. Random sampling also performed well, consistent with the benefit of distributed spatial sampling, while custom masks achieved competitive performance at the

cost of manual design. Spatial averaging provided a useful trade-off at moderate compression but degraded under stronger spatial aggregation.

TABLE II: Validation accuracy (%) for compression methods. Numbers in parentheses indicate the retained input dimensionality.

Method	Size 1	Size 2	Size 3
FPS	86.22 (400)	75.16 (200)	58.98 (100)
Random		80.60 (300)	
PCA-rank		84.24 (400)	
H-Stripes		83.43 (288)	
ROI		78.39 (300)	
Avg	82.08 (302)	—	55.00 (151)

Figure 5 shows the confusion matrix for the selected FPS configuration on the full 17-class task. Several classes exhibit strong diagonal dominance, whereas confusions remain between objects producing similar spatial pressure patterns. These results highlight the difficulty of discriminating morphologically similar objects from tactile information alone.

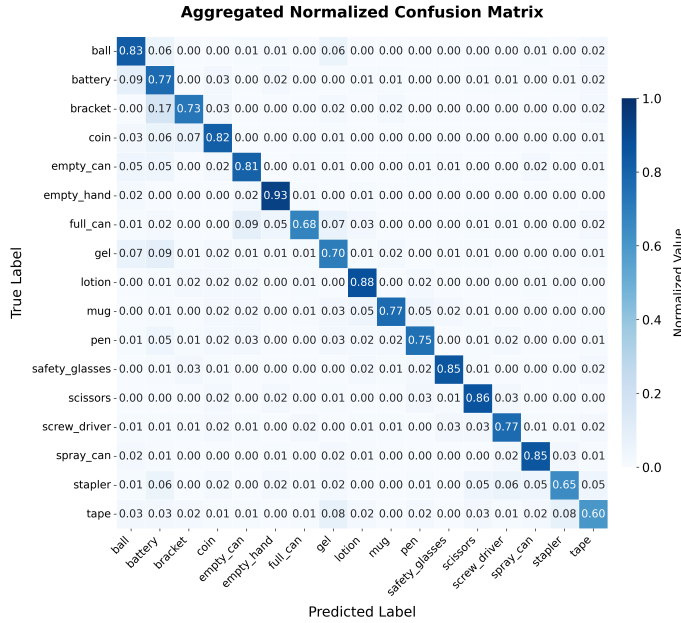


Fig. 5: Normalized confusion matrix for the 17-class task with 256 input pixels.

To assess inter-session generalization, the final model was evaluated across all five acquisition sessions using leave-one-session-out cross-validation. The SNN results were compared with the frame-based CNN results previously reported in [22] under the corresponding evaluation protocols (Fig. 6). While the CNN achieved higher Top-1 accuracy under random splits, the SNN achieved competitive Top-3 performance. Under inter-session evaluation, the SNN achieved $57.45 \pm 11.11\%$ Top-1 and $87.55 \pm 3.75\%$ Top-3 accuracy, compared with $37.75 \pm 10.04\%$ and $63.41 \pm 11.99\%$, respectively, for the previously reported CNN. These results indicate improved

generalization of the spike-based representation across acquisition sessions.

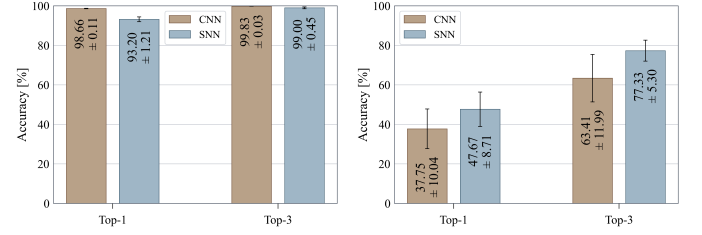


Fig. 6: Classification accuracy for random-split (left) and inter-session (right) evaluation. Inter-session results are obtained using leave-one-session-out evaluation across all five acquisition sessions.

A. On-Chip Proof-of-Concept Results

A hardware-constrained experiment assessed deployment feasibility on DYNAP-SE [21]. The task was reduced to 10 well-separated classes and FPS compression to $N = 128$ inputs to accommodate hardware resource and connectivity constraints. The mapped network comprised 128 input generators, 384 hidden neurons, and 10 outputs, with a maximum fan-in of 64 synapses per neuron, reducing synaptic density by more than 75% relative to the dense configuration.

As shown in Table III, the hardware-constrained simulation achieved $42.04 \pm 8.78\%$ Top-1 and $71.49 \pm 7.02\%$ Top-3 accuracy, compared with $30.98 \pm 8.33\%$ and $58.55 \pm 7.43\%$ on DYNAP-SE. The simulation-to-hardware gap is attributed to mixed-signal non-idealities, including device mismatch, bias variability, and noise, together with limited on-chip parameter calibration.

These results demonstrate deployment feasibility under real hardware constraints, while the on-chip experiment remains a proof of concept rather than an application-ready implementation.

TABLE III: Simulation and on-chip performance for the hardware-constrained 10-class inter-session task.

Implementation	Top-1	Top-3
Simulation	$42.04 \pm 8.78\%$	$71.49 \pm 7.02\%$
DYNAP-SE chip	$30.98 \pm 8.33\%$	$58.55 \pm 7.43\%$

IV. CONCLUSION

This work presents a spike-based tactile object classification pipeline using SNNs, combining event-driven encoding with tactile input compression to transform high-dimensional tactile measurements into compact representations suitable for neuromorphic processing. Through the systematic evaluation of multiple compression strategies, we identify design trade-offs between input dimensionality and classification performance while demonstrating the feasibility of end-to-end deployment on mixed-signal neuromorphic hardware. These findings provide practical guidelines for the design of future neuromorphic tactile systems for robotic and prosthetic edge applications.

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